

Less Traffic, Higher Tolls: Optimal Congestion Pricing under Remote Work and Traveler Heterogeneity

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Abstract

This paper studies how a persistent shift toward remote work changes optimal freeway congestion prices. I combine hourly station traffic from the California Performance Measurement System with tract-level commuting patterns, work-from-home shares, industry composition, population, and wages for five large California counties. Station traffic is allocated to residential tracts using fixed pre-pandemic commuting-route exposure, producing an estimated tract-level freeway VMT measure that aggregates exactly to observed PeMS VMT. A shift-share design based on predetermined local industry shares and national leave-California-out changes in industry remote work implies that a 10 percentage point increase in working from home reduces tract freeway VMT by 1.43 percent. A capacity-normalized BPR technology estimated by Poisson pseudo-maximum likelihood yields a congestion curvature of 3.27. These inputs produce three main pricing findings. First, holding the value of time fixed, the demand shift lowers average optimal tolls by 2.4–4.7 percent. Second, allowing the value of time to move with local wages reverses the result: optimal tolls rise by 8.0–11.4 percent. Third, income heterogeneity changes the meaning of distributional comparisons. Lower-income tracts supply 77–84 percent of modeled traffic and bear most of the marginal external damages in the common Pigouvian calculation. Under a constrained policy that imposes the same time-equivalent charge on both groups, however, the monetary rate paid by higher-income travelers is 1.6–2.0 times larger because their value of time is higher. Distributional conclusions therefore depend on whether the object of interest is a group's role in and exposure to external delay or the monetary rate it pays.

Keywords: Congestion pricing, remote work, vehicle miles traveled, value of time, shift-share instruments, distributional incidence

JEL Classification: R40, R41, D62, H23

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1 Introduction

Traffic congestion is a textbook example of an externality. A driver entering a crowded road considers the time she expects to spend traveling but not the delay imposed on every other road user. The resulting private equilibrium contains too much traffic relative to the social optimum. Beginning with [Pigou \(1920\)](#) and [Vickrey \(1969\)](#), the standard prescription has therefore been to charge each vehicle the marginal external congestion cost it creates. The bottleneck and network-pricing literature formalizes how the efficient charge depends on departure-time choice, elastic demand, and network constraints ([Arnott, de Palma, & Lindsey, 1993](#); [Newbery, 1989](#); [Small, Verhoef, & Lindsey, 2024](#); [Verhoef, 2002](#)).

Empirical congestion-pricing studies show why both the road technology and the behavioral response matter for policy. Evidence from London and Milan demonstrates that pricing changes traffic while also inducing substitution across times, routes, and modes ([Gibson & Carnovale, 2015](#); [Leape, 2006](#)). [Yang, Purevjav, and Li \(2020\)](#) use a driving restriction in Beijing to identify the density–speed relationship and calculate location- and time-specific congestion charges. [Kreindler \(2024\)](#) combines a field experiment with an equilibrium model of peak-hour travel in Bangalore and shows that severe observed congestion need not imply large welfare gains from pricing when marginal external delay is modest. Related evidence from California freeway bottlenecks cautions against interpreting every decline in speed as a decline in physical throughput ([Anderson & Davis, 2020](#)). This paper contributes to that empirical literature by estimating a capacity-normalized congestion technology from a large county-hour detector panel and combining it with a demand shift that is observed at a finer residential geography.

This calculation is often presented as an engineering exercise, but its monetary component is economic. The same physical minute of delay produces a larger dollar externality when travelers have a higher value of time. Values of time vary considerably across people and

settings ([Bento, Roth, & Waxman, 2024](#); [Lam & Small, 2001](#); [Small, 2012](#); [Small, Winston, & Yan, 2005](#)), and this heterogeneity affects both the efficiency and distributional incidence of road pricing ([Small & Yan, 2001](#); [van den Berg & Verhoef, 2011](#); [Verhoef & Small, 2004](#)). Distributional effects also depend on baseline travel patterns, available alternatives, and the use of toll revenue rather than on income alone ([Eliasson & Mattsson, 2006](#)). [Hall \(2018\)](#) further shows that time-varying prices on only part of a congested highway can, under particular sorting conditions, benefit all road users even before revenue is recycled. A policy calculation that updates traffic conditions while holding the value of time fixed can therefore misstate both the level and the incidence of the efficient charge. I make this distinction operational by separating the common monetary Pigouvian toll on a shared road from a constrained schedule that assigns the same time-equivalent charge, but different dollar-per-mile rates, to travelers with different values of time.

I study this interaction using the shift toward remote work following the COVID-19 pandemic. The empirical setting consists of Los Angeles, Orange, San Diego, San Francisco, and Santa Clara counties. Hourly freeway conditions come from the California Performance Measurement System (PeMS). Demographic characteristics, industry composition, remote work, and wages come from the American Community Survey (ACS), while origin–destination commuting patterns come from the Longitudinal Employer-Household Dynamics Origin-Destination Employment Statistics (LODES). The analysis compares 2018–2019 with 2021–2022, omitting the transition year 2020 from the tax comparison.

The paper is also related to the literature on telecommuting, urban form, and travel demand. [Safirova \(2002\)](#) formalizes the trade-off between lower commuting congestion and weaker workplace agglomeration. Quantitative urban models subsequently show that persistent remote work can change residential and employment locations as well as congestion, including within the Los Angeles metropolitan area ([Delventhal, Kwon, & Parkhomenko, 2022](#)). The feasibility of working from home varies sharply across occupations and indus-

tries ([Dingel & Neiman, 2020](#)), creating uneven local exposure to the expansion of remote work. Survey evidence indicates that the increase in working from home remained far above its pre-pandemic level ([Barrero, Bloom, & Davis, 2023](#)), while recent work emphasizes that temporary commuting disruptions can produce persistent changes in remote-work equilibria, especially in large cities ([Monte, Porcher, & Rossi-Hansberg, 2026](#)). Transportation evidence also finds that congestion relief from working from home varies substantially across locations and times of day ([Loo & Huang, 2022](#)). I complement this literature by estimating how within-county differences in remote-work exposure change freeway VMT and then carrying that demand response into a congestion-pricing calculation.

Finally, the demand analysis relates to work showing that vehicle travel responds endogenously to road capacity and generalized driving cost. [Duranton and Turner \(2011\)](#) document a near-proportional long-run response of VMT to highway capacity across U.S. cities, and [Hymel, Small, and Van Dender \(2010\)](#) estimate induced-demand and rebound channels jointly with congestion. Rather than using infrastructure expansion or fuel cost as the demand shifter, I use the pandemic-era change in the feasibility and incidence of working from home. Fixed pre-pandemic commuting-route exposure then supplies the cross-sectional allocation needed to connect observed freeway traffic to residential tracts.

The first empirical contribution is a tract-level measure of freeway VMT. PeMS observes station traffic but does not assign that traffic to residential census tracts. LODES identifies where workers live and work but measures jobs and commuting relationships rather than realized trips. I combine their strengths. For each PeMS station, I construct fixed pre-pandemic exposure weights from the number of commuters whose modeled freeway route passes the station. I then allocate the station’s observed yearly VMT across origin tracts using those weights. The resulting measure is an estimate of the freeway VMT associated with each tract, not a direct count of vehicles from that tract. Its central accounting property is exact aggregation: summing the allocated outcome over tracts reproduces observed PeMS VMT in

the county.

The second contribution is a tract-level demand design. The work-from-home share may respond to local labor-market conditions, residential selection, and unobserved transportation shocks. I instrument it with 2019 tract industry shares interacted with national, leave-California-out changes in industry-specific work-from-home rates. With tract and county-by-year fixed effects, identification comes from differential pre-pandemic industry exposure among tracts in the same county and year. The preferred estimate implies that a 10 percentage point increase in working from home reduces allocated tract freeway VMT by 1.43 percent.

The third contribution is an internally consistent congestion-pricing exercise. I estimate a capacity-normalized BPR curve rather than absorbing capacity into an uninterpretable scale parameter. Station-specific free-flow speeds are obtained from uncongested hours, while engineering capacity uses the observed number of lanes, PeMS station coverage length, and standard passenger-car-equivalent benchmarks. A Poisson pseudo-maximum-likelihood specification accommodates the large number of observations with zero measured delay. The preferred estimate has county-specific scale parameters and a common congestion curvature of 3.27.

The pricing exercise delivers three main findings. First, when the value of time is fixed at its pre-period level, the demand shift reduces average optimal taxes in all five counties, by 2.4 to 4.7 percent. Second, with contemporaneous wages, optimal taxes instead rise by 8.0 to 11.4 percent. An exact decomposition explains this reversal: lower congestion reduces the tax by 7.3 to 9.4 percent of its pre-period value, while the higher value of time raises it by 17.0 to 18.8 percent. The wage component is larger, so the net monetary tax rises.

The third main finding concerns income heterogeneity and has a distinct policy interpretation. I divide tracts at the household-weighted 80th percentile of each county's 2018 median household income and re-estimate the demand response separately for the two groups. I use each group's baseline freeway-VMT share to recover its 2018 hourly flow, calibrate a

group-specific demand intercept, and solve both demands simultaneously because they create congestion on the same network. The unconstrained first best remains a common monetary Pigouvian toll: the identity of the entering driver does not change the physical delay imposed on other road users. Its marginal damages can nevertheless be decomposed across the groups exposed to that delay. Lower-income tracts supply 77–84 percent of equilibrium flow and bear most of this damage component. By contrast, under a constrained policy that gives both groups the same time-equivalent charge, higher-income travelers pay dollar-per-mile rates 1.6–2.0 times larger because their value of time is higher. Thus, the same group can represent most traffic and bear most external-delay damage while facing the lower monetary rate under a time-equivalent schedule. The distributional conclusion depends on the object being measured: responsibility for and exposure to delay, or the monetary rate paid. Keeping these objects separate is a central contribution of the paper.

A complementary time-of-day analysis shows that the first two pricing findings are not artifacts of temporal aggregation. I re-estimate the congestion technology separately for the morning peak, afternoon peak, and off-peak hours. Taxes are largest around the commuting peaks and effectively zero in some off-peak hours; the model never predicts a subsidy. When wages are allowed to change, the post-period schedule generally lies above the pre-period schedule.

The remainder of the paper is organized as follows. Section 2 develops the model. Section 3 describes the data and the tract allocation. Section 4 presents the congestion and demand specifications. Section 5 reports the main pricing results and distributional analysis. Section 6 concludes. The appendices provide derivations and implementation details.

2 Model

Computing an optimal congestion charge requires three objects: a technology that maps traffic into travel time, a demand curve that maps generalized cost into freeway use, and a value of time that converts delay into dollars. County is indexed by c , hour of the weekly profile by h , calendar year by t , and flow by F in passenger-car-equivalent vehicle miles per hour.

2.1 Congestion technology and travel costs

Let S_c^0 denote free-flow speed and K_c the engineering capacity of the observed freeway network. Following the functional form introduced by [U.S. Bureau of Public Roads \(1964\)](#), the BPR travel-time function per mile is

$$T_c(F) = \frac{1}{S_c^0} \left[1 + \alpha_c \left(\frac{F}{K_c} \right)^\beta \right], \quad (1)$$

where $\alpha_c > 0$ is a county-specific scale parameter and $\beta > 0$ controls curvature. The ratio F/K_c is dimensionless, and $T_c(F)$ is measured in hours per mile. Differentiation gives

$$T'_c(F) = \frac{\alpha_c \beta}{S_c^0 K_c} \left(\frac{F}{K_c} \right)^{\beta-1}. \quad (2)$$

Let VOT_{ct} denote dollars per hour. The private marginal cost of one vehicle mile is the time cost borne by its driver,

$$PMC_{cht}(F) = VOT_{ct} T_c(F) = \frac{VOT_{ct}}{S_c^0} \left[1 + \alpha_c \left(\frac{F}{K_c} \right)^\beta \right]. \quad (3)$$

Total travel-time cost is $F VOT_{ct} T_c(F)$. Its derivative with respect to flow yields marginal

social cost,

$$MSC_{cht}(F) = PMC_{cht}(F) + MECC_{cht}(F), \quad (4)$$

$$MECC_{cht}(F) = VOT_{ct} FT'_c(F) = \frac{VOT_{ct}}{S_c^0} \alpha_c \beta \left(\frac{F}{K_c} \right)^\beta. \quad (5)$$

Thus $MECC$ is measured in dollars per mile. Appendix A derives Equations (4) and (5) from the driver's problem and the planner's problem.

I set the value of time to one-half of the local hourly wage,

$$VOT_{ct} = \theta w_{ct}, \quad \theta = 0.5, \quad (6)$$

following the conventional benchmark summarized by [Small \(2012\)](#). The factor is held fixed across all policy exercises; what changes is whether the underlying wage is fixed at its 2018 value or allowed to vary by county and year.

2.2 Demand, equilibrium, and the Pigouvian tax

Let p be the generalized monetary price perceived by a driver. Freeway demand is constant elasticity,

$$F_{cht} = A_{cht} p_{cht}^{-\eta}, \quad \eta = 0.1, \quad (7)$$

where A_{cht} is a positive demand shifter and $-\eta$ is the elasticity of freeway use with respect to generalized monetary cost. The empirical remote-work design determines changes in A_{cht} ; it does not estimate η . Following [Yang et al. \(2020\)](#), who also impose an elasticity of -0.1 in their congestion-pricing counterfactuals, I set $\eta = 0.1$ throughout rather than estimating it.

With a monetary toll τ_{cht} , private equilibrium satisfies

$$F_{cht} = A_{cht} [PMC_{cht}(F_{cht}) + \tau_{cht}]^{-\eta}. \quad (8)$$

The planner equates inverse demand with marginal social cost. Efficient flow F_{cht}^* therefore solves

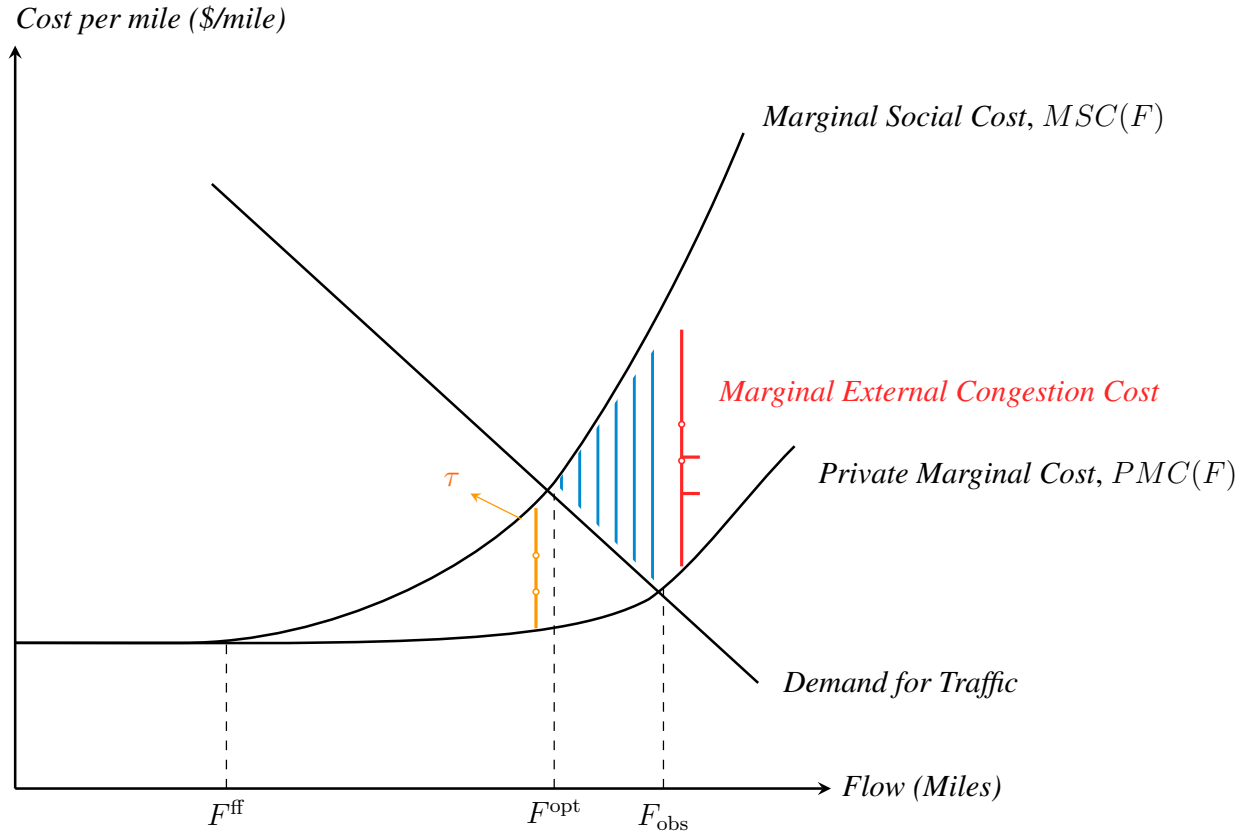
$$F_{cht}^* = A_{cht} [MSC_{cht}(F_{cht}^*)]^{-\eta}, \quad (9)$$

and is decentralized by the Pigouvian tax

$$\tau_{cht}^* = MECC_{cht}(F_{cht}^*). \quad (10)$$

The tax is not chosen to make traffic as small as possible. It is the value of the delay imposed on other travelers at the efficient flow. Raising the tax beyond Equation (10) would remove trips whose willingness to pay exceeds their social cost and would lower welfare. Figure 1 illustrates this result. Without a toll, demand intersects private marginal cost at F_{obs} . Marginal social cost lies above private marginal cost by the marginal external congestion cost, so the efficient intersection occurs at the lower flow F^{opt} . The vertical gap between the two cost curves at F^{opt} is the optimal toll, and the hatched wedge is the deadweight loss from unpriced congestion. Appendix B provides the formal first-order condition.

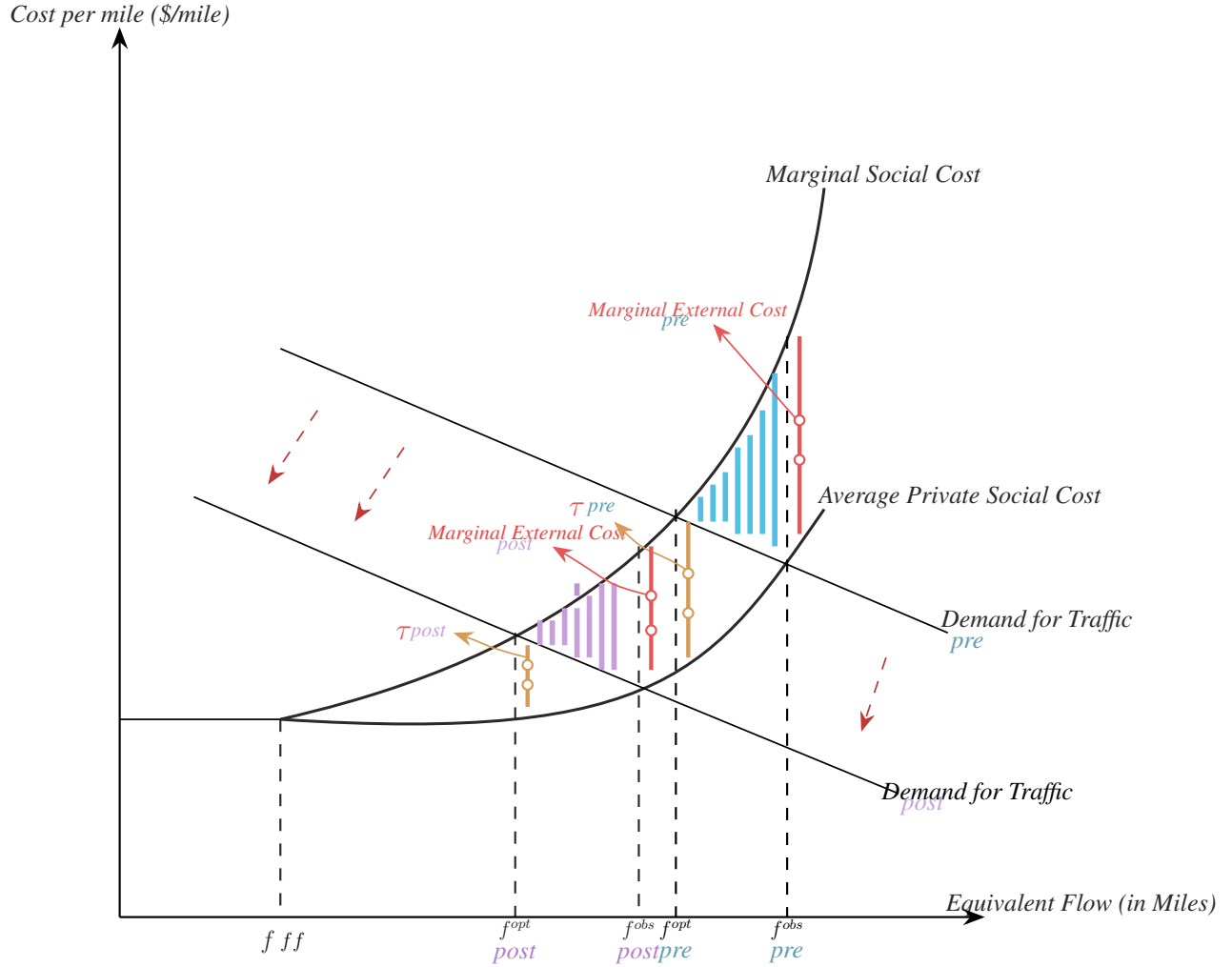
Figure 1. Private and Marginal Social Costs of Congestion



Notes: The unpriced flow F_{obs} occurs where demand intersects private marginal cost. Efficient flow F^{opt} occurs where demand intersects marginal social cost. The optimal per-mile toll equals marginal external congestion cost at the efficient flow. The hatched region represents the deadweight loss from unpriced congestion.

Figure 2 applies the same logic before and after a decline in travel demand. Holding the congestion technology and the value of time fixed, the inward demand shift moves both the unpriced and efficient intersections to lower flows. Because marginal external delay is lower at the new efficient flow, the optimal toll falls. This is the mechanism isolated by the fixed-value-of-time counterfactual; the empirical analysis subsequently allows wages, and therefore the monetary value attached to delay, to change as well.

Figure 2. Optimal Congestion Pricing Before and After a Demand Shift



Notes: A downward demand shift reduces unpriced and efficient flow. Holding the congestion technology and value of time fixed, the marginal external cost and optimal tax fall. The empirical exercise separately allows the value of time to change.

2.3 Toll incidence and congestion-cost offset

Suppressing subscripts, write the perceived price as $p(\tau) = PMC(F(\tau)) + \tau$. Taking logs of the private-equilibrium condition gives

$$\ln F = \ln A - \eta \ln[PMC(F) + \tau]. \quad (11)$$

Differentiating both sides with respect to the toll,

$$\frac{1}{F} \frac{dF}{d\tau} = -\frac{\eta}{PMC(F) + \tau} \left[PMC'(F) \frac{dF}{d\tau} + 1 \right]. \quad (12)$$

Collecting the terms that contain $dF/d\tau$ gives

$$\frac{dF}{d\tau} = -\frac{\eta F}{PMC(F) + \tau + \eta F PMC'(F)} < 0. \quad (13)$$

The toll reduces flow, which improves traffic conditions and lowers private time cost:

$$\frac{dPMC}{d\tau} = -\frac{\eta F PMC'(F)}{PMC(F) + \tau + \eta F PMC'(F)} < 0. \quad (14)$$

Accordingly,

$$0 < \frac{dp}{d\tau} = \frac{PMC(F) + \tau}{PMC(F) + \tau + \eta F PMC'(F)} \leq 1. \quad (15)$$

The perceived price rises less than dollar-for-dollar because part of the charge is returned to drivers through lower congestion. This does not imply that an arbitrarily large tax is efficient. Equation (15) is an incidence result at any feasible tax; Equation (10) is the welfare condition selecting the tax. At the optimum,

$$\tau^* = F^* PMC'(F^*) = MECC(F^*), \quad (16)$$

so the pass-through expression becomes

$$\begin{aligned} \left. \frac{dp}{d\tau} \right|_{\tau=\tau^*} &= \frac{PMC(F^*) + MECC(F^*)}{PMC(F^*) + MECC(F^*) + \eta MECC(F^*)} \\ &= \frac{MSC(F^*)}{MSC(F^*) + \eta MECC(F^*)} \in (0, 1). \end{aligned} \quad (17)$$

Thus the optimal toll is determined by the marginal external cost at the efficient flow, while Equation (17) describes how a marginal change in that toll is divided between a higher perceived price and lower congestion cost. This distinction matters for interpretation: a one-dollar increase in the posted toll raises the driver's generalized price by less than one dollar because the induced reduction in traffic returns part of the charge as time savings. Pass-through therefore describes the incidence of a marginal tax change, not whether the tax itself is welfare maximizing.

2.4 Aggregation and decomposition

I solve Equations (9)–(10) for each county, year, and hour of the 168-hour weekly profile. For county c and period r , the reported tax and value of time are the optimal-flow-weighted means

$$\bar{\tau}_{cr} = \frac{\sum_{t \in r} \sum_{h \in \mathcal{H}} \tau_{cht}^* F_{cht}^*}{\sum_{t \in r} \sum_{h \in \mathcal{H}} F_{cht}^*}, \quad \overline{VOT}_{cr} = \frac{\sum_{t \in r} \sum_{h \in \mathcal{H}} VOT_{cht} F_{cht}^*}{\sum_{t \in r} \sum_{h \in \mathcal{H}} F_{cht}^*}, \quad (18)$$

where $\mathcal{H}$ is the set of weekly hours and $r \in \{pre, post\}$. Thus $\bar{\tau}_{cr}$ is the expected tax per vehicle mile at the efficient allocation, rather than a simple unweighted average of hourly rates. I define the effective period marginal external delay index as

$$g_{cr} \equiv \frac{\bar{\tau}_{cr}}{\overline{VOT}_{cr}}, \quad \bar{\tau}_{cr} = \overline{VOT}_{cr} g_{cr}. \quad (19)$$

The index g_{cr} is measured in hours per mile. When the value of time is constant within a period, it reduces to the optimal-flow-weighted mean of the cell-level marginal external delay $F_{cht}^* T'_c(F_{cht}^*)$ derived in Appendix A. When values of time vary across years, Equation (19) is the effective tax-to-value-of-time ratio used in the exact accounting decomposition. Sup-

pressing the county subscript and adding and subtracting $\overline{VOT}_{pre}g_{post}$ gives

$$\Delta\bar{\tau} = \underbrace{\overline{VOT}_{pre}(g_{post} - g_{pre})}_{\text{congestion component}} + \underbrace{g_{post}(\overline{VOT}_{post} - \overline{VOT}_{pre})}_{\text{value-of-time component}}. \quad (20)$$

The first term asks how the tax changes when physical marginal delay changes but the value of time does not. The second prices post-period delay using the change in the monetary value of time.

3 Data

3.1 PeMS traffic and the observed freeway network

PeMS records hourly flow, VMT, speed, lane counts, and station location for California freeways. I use mainline and HOV stations in the five study counties. Passenger-car-equivalent flow is

$$F_{cht} = VMT_{cht}^{passenger} + 1.5 VMT_{cht}^{truck}. \quad (21)$$

The factor 1.5 converts truck VMT into passenger-car equivalents. It is a transparent engineering benchmark rather than an estimated parameter.

Free-flow speed is estimated at the station level using low-flow observations in 2018 and 2019, requiring speeds between 50 and 90 miles per hour and adequate detector coverage. Mainline capacity per lane follows the standard engineering schedule $1,700 + 10S_s^0$ passenger cars per hour per lane when $S_s^0 \leq 70$ mph, capped at 2,400 above 70 mph. HOV capacity is set to 1,650 passenger cars per hour per lane.¹ Station capacity is multiplied by the number of lanes and the PeMS coverage length; county capacity K_c is the sum over stations. Table 1 reports the resulting network inputs.

¹The capacity benchmarks follow the Federal Highway Administration's HPMS guidance and Freeway Management Handbook.

Table 1. Estimated Congestion Technology and Engineering Inputs

County	$\hat{\alpha}_c$	S_c^0 (mph)	K_c (million PCE-VMT/h)	Stations	Lane-miles
Los Angeles	2.583	66.85	10.407	2,623	4,562.9
Orange	1.649	67.37	4.402	1,667	1,950.7
San Diego	2.350	67.19	5.533	891	2,388.5
San Francisco	3.260	66.82	0.406	62	171.6
Santa Clara	2.720	66.26	2.497	554	1,058.6

Notes: The preferred PPML specification estimates county-specific scale parameters and a common curvature parameter, $\hat{\beta} = 3.270$ (HC1 standard error 0.019), using 175,180 county-hour observations. S_c^0 is the capacity-mile-weighted harmonic mean of station free-flow speeds. K_c sums station engineering capacity times PeMS station coverage length over mainline and HOV detectors. Capacity is measured in passenger-car-equivalent VMT per hour (PCE-VMT/h); trucks receive a factor of 1.5.

These capacity measures describe the freeway mileage represented by the detector network, not the administrative capacity of every road in the county. This alignment is important: the numerator F_{cht} and denominator K_c cover the same observed network.

3.2 Allocating station VMT to residential tracts

PeMS does not identify the residential origin of a vehicle. I therefore allocate observed station VMT to origin census tracts using pre-pandemic LODES commuting pairs and modeled freeway routes:

$$\widehat{VMT}_{it} = \sum_{s \in \mathcal{S}_{c(i)}} \left[\frac{\sum_j \bar{N}_{ij}^{pre} A_{ijs}}{\sum_{k \in \mathcal{I}_{c(i)}} \sum_j \bar{N}_{kj}^{pre} A_{kjs}} \right] \left[\sum_{h \in t} Flow_{sh} L_s \right]. \quad (22)$$

Here i indexes the residential origin tract and j the workplace destination tract. The commuting count $\bar{N}_{ij}^{pre}$ is the average number of LODES workers traveling from i to j before the pandemic. The indicator A_{ijs} equals one when the modeled freeway route between the centroids of i and j traverses the freeway edge represented by PeMS station s . The set $\mathcal{S}_{c(i)}$ contains the stations in the county of origin tract i .

The index k runs over every candidate residential origin tract in $\mathcal{I}_{c(i)}$, including tract i . Thus the denominator is the total pre-pandemic routed-worker exposure to station s across all origin tracts in that county, and the ratio in brackets is the share assigned to tract i . The indices i and j already identify a unique commuting pair, so the commuting count requires no additional county index.

In the second bracket, $Flow_{sh}$ is observed traffic flow at station s in hour h , L_s is the freeway coverage length represented by that station in miles, and $\sum_{h \in t} Flow_{sh} L_s$ is annual station VMT. The allocation therefore does not weight every commuting pair equally. Routes that use more of the observed freeway network encounter more station exposure, while each station contributes only the VMT associated with its own coverage length. The pre-period route weights are fixed, so the pandemic movement in $\widehat{VMT}_{it}$ comes from observed PeMS traffic rather than from contemporaneous LODES commuting counts.

For every matched station, the bracketed allocation shares sum to one across origin tracts. Consequently,

$$\sum_{i \in c} \widehat{VMT}_{it} = \sum_{s \in c} VMT_{st}^{PeMS}. \quad (23)$$

Stations without a route match are allocated using fixed baseline origin-worker shares, so the county accounting identity is preserved. Equation (23) is an accounting reconciliation rather than an external validation test. The cross-sectional allocation comes from LODES and modeled route exposure; the aggregate time-series movement comes from observed PeMS traffic.

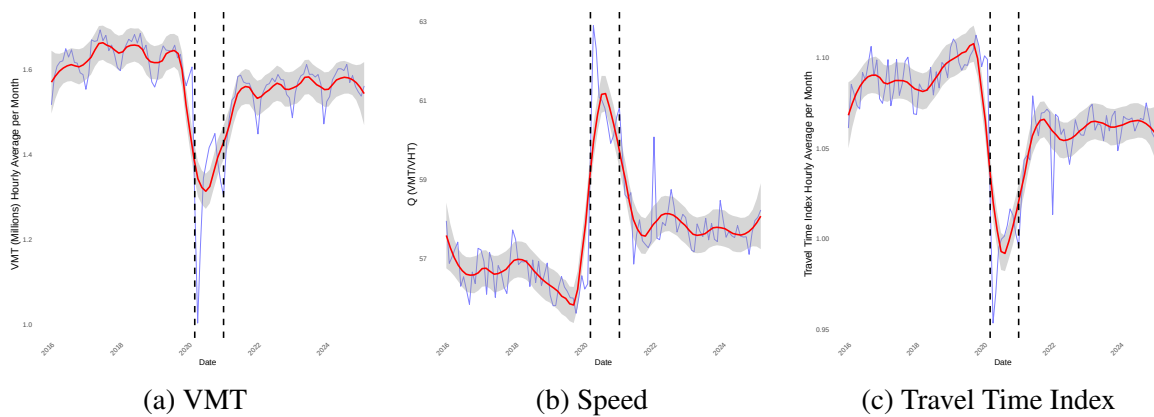
3.3 Remote work, industry exposure, and wages

ACS five-year estimates provide tract population, work-from-home shares, unemployment, median household income, poverty, college attainment, vehicle availability, civilian employment by industry, and earnings. LODES provides the origin–destination commuting counts

used in Equation (22). Section 4 introduces the shift–share design after presenting the demand equation and explains how the industry variables identify differential exposure to national changes in remote work.

Figure 3 shows the county-level traffic series. VMT fell sharply in 2020, average speed rose, and TTI fell. Traffic subsequently recovered in part, but the persistent change in commuting behavior motivates separating the pre-period 2018–2019 from the post-period 2021–2022.

Figure 3. Freeway Traffic and Congestion



Notes: County traffic measures are constructed from hourly PeMS data. The dashed vertical marker denotes the onset of the pandemic.

Appendix Figure 11 shows the corresponding increase in county work-from-home shares, while Appendix Figure 12 documents how the changes in VMT and congestion vary over the 24-hour day. Together with Figure 3, these figures show that the pandemic shift affected both the level and the timing of freeway use.

4 Estimation

4.1 Congestion technology

Let Q_{cht} denote observed average speed. Proportional delay relative to free flow is

$$D_{cht} = \max \left\{ \frac{S_c^0}{Q_{cht}} - 1, 0 \right\}. \quad (24)$$

The truncation at zero treats speeds above the estimated free-flow benchmark as no measured congestion rather than negative delay. The preferred conditional-mean specification is

$$E[D_{cht} \mid F_{cht}, c] = \alpha_c \left(\frac{F_{cht}}{K_c} \right)^\beta, \quad (25)$$

estimated by Poisson pseudo-maximum likelihood. PPML is useful here because D_{cht} is non-negative and frequently zero; log OLS would discard zero-delay observations and mechanically select the congested part of the sample. The estimator is used as a quasi-likelihood for the conditional mean and does not require D_{cht} to be a count.

The preferred specification contains county-specific α_c and one common β . It uses 175,180 county-hour observations in 2018, 2019, 2021, and 2022. The estimated curvature is $\hat{\beta} = 3.270$ with an HC1 standard error of 0.019. Table 1 reports the county scale estimates. Because flow and speed are jointly determined within the hour, I interpret Equation (25) as the congestion curve used for structural calibration rather than as a fully causal supply estimate. This distinction follows the identification concerns emphasized by [Anderson and Davis \(2020\)](#) and the excluded-demand-shifter approaches in [Yang et al. \(2020\)](#) and [Kreindler \(2024\)](#). Weather, incidents, and predetermined traffic patterns provide promising sources of additional identification.

4.2 Tract-level demand

The empirical demand equation is

$$\ln \widehat{VMT}_{it} = \xi_i + \delta_{c(i)t} + \rho WFH_{it} + \gamma \ln Population_{it} + X'_{it}\pi + \varepsilon_{it}, \quad (26)$$

where WFH_{it} is measured in percentage points, ξ_i are tract fixed effects, and $\delta_{c(i)t}$ are county-by-year fixed effects. County-by-year effects absorb fuel prices, county labor-market conditions, policies, and any other shock common to tracts in the same county and year. The preferred specification sets X_{it} empty; a robustness specification adds unemployment, household income, poverty, college attainment, and vehicle availability.

Because remote work is endogenous, I instrument it with

$$Z_{it} = 100 \sum_q s_{iq,2019} \left(WFH_{q,t}^{US\backslash CA} - WFH_{q,2019}^{US\backslash CA} \right), \quad (27)$$

where $s_{iq,2019}$ is the predetermined employment share of industry q in tract i , and the term in parentheses is the national change in that industry's work-from-home rate calculated outside California. The identifying assumption is that, conditional on tract and county-by-year fixed effects, predetermined differences in local industry exposure affect tract freeway VMT through their effect on remote work rather than through a separate tract-specific channel aligned with the national industry shock.

Equation (27) is an exposure design in the terminology of [Goldsmith-Pinkham, Sorkin, and Swift \(2020\)](#) and a shock-based shift–share design in the terminology of [Borusyak, Hull, and Jaravel \(2022\)](#). The empirical argument relies on the national industry shocks being as-good-as-random with respect to the appropriately weighted tract residuals, while the 2019 shares determine differential exposure.

Table 2 presents the estimates. The preferred coefficient is -0.001441 , so a 10 percentage

point increase in remote work changes VMT by $100[\exp(10\hat{\rho}) - 1] = -1.43$ percent. Adding demographic controls gives a similar effect of -1.16 percent. The clustered first-stage Wald statistics are large, and standard errors are two-way clustered by tract and year.

Table 2. Remote Work and Tract-Level Freeway VMT

	(1) Preferred IV	(2) IV with controls
Work-from-home share (percentage points)	-0.001441^{**} (0.000582)	-0.001167^* (0.000486)
Effect of a 10 p.p. increase (percent)	-1.43	-1.16
First-stage coefficient	3.606	3.622
First-stage clustered Wald statistic	1,006.6	744.9
Observations	25,088	24,975
Tract fixed effects	Yes	Yes
County-by-year fixed effects	Yes	Yes
Log population	Yes	Yes
Additional demographic controls	No	Yes

Notes: The dependent variable is log tract-level freeway VMT. The work-from-home share is instrumented with 2019 tract industry shares interacted with national, leave-California-out changes in industry-specific work-from-home rates. Column (2) additionally controls for unemployment, median household income, poverty, college attainment, and the share of households without a vehicle. Standard errors in parentheses are two-way clustered by tract and year. The first-stage statistic is the clustered Wald statistic for the excluded instrument. Because the panel contains seven years, the reported inference should be interpreted cautiously; exposure-robust shift-share inference is an important next step. $^*p < 0.10$, $^{**}p < 0.05$.

4.3 Mapping the demand estimate into hourly policy simulations

For each tract, define the predicted demand multiplier relative to 2018,

$$m_{it} = \exp \{ \hat{\rho}(WFH_{it} - WFH_{i,2018}) + \hat{\gamma}[\ln Population_{it} - \ln Population_{i,2018}] \}. \quad (28)$$

I aggregate these changes using fixed 2018 tract-VMT weights,

$$M_{ct} = \sum_{i \in c} \nu_{i,2018} m_{it}, \quad \nu_{i,2018} = \frac{\widehat{VMT}_{i,2018}}{\sum_{k \in c} \widehat{VMT}_{k,2018}}. \quad (29)$$

The 2018 demand intercept is calibrated separately for every county, day of week, and hour so that observed baseline flow satisfies private equilibrium:

$$A_{cdh,2018} = F_{cdh,2018}^{obs} [PMC_{c,2018}(F_{cdh,2018}^{obs})]^\eta. \quad (30)$$

Subsequent demand is $A_{cdht} = A_{cdh,2018}M_{ct}$. I then solve Equation (9) numerically and evaluate the tax in Equation (10). This procedure preserves the 168-hour shape of baseline traffic while introducing the causally estimated remote-work and population shifts.

5 Results

5.1 Optimal taxes with fixed and contemporaneous values of time

Table 3 reports optimal-flow-weighted taxes. Under a fixed 2018 value of time, post-period taxes fall in every county. The reductions range from 2.39 percent in San Diego to 4.75 percent in San Francisco. These estimates isolate the physical effect of the demand shift: lower efficient traffic reduces marginal external delay.

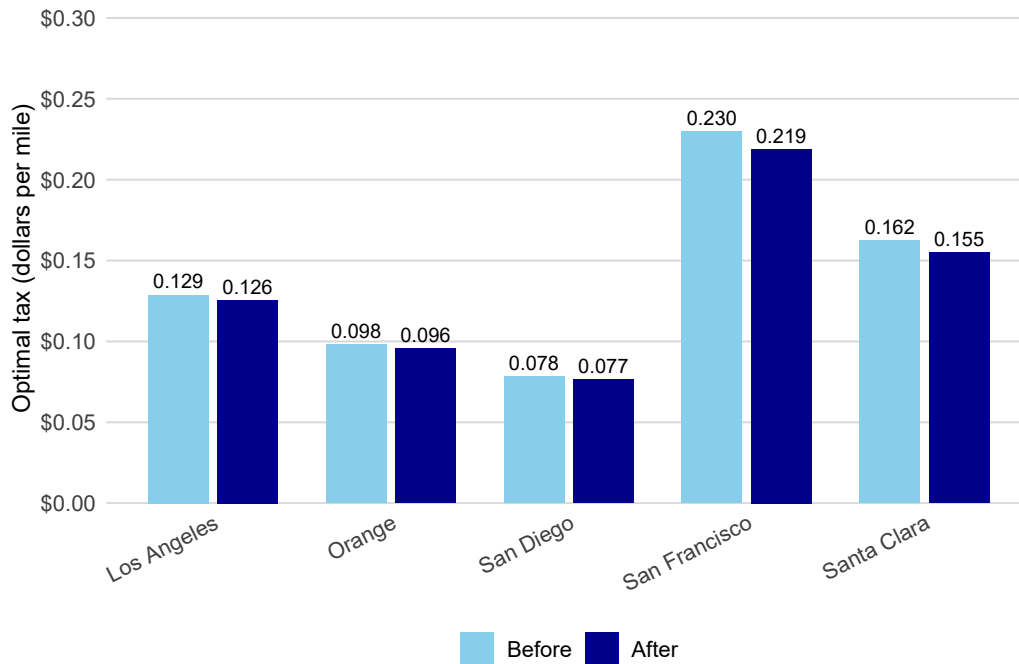
Table 3. Optimal Congestion Taxes Before and After the Demand Shift

County	Fixed pre-period value of time			Contemporaneous value of time		
	Pre	Post	Change (%)	Pre	Post	Change (%)
Los Angeles	0.129	0.126	−2.52	0.130	0.145	11.38
Orange	0.098	0.096	−2.78	0.098	0.109	10.84
San Diego	0.078	0.077	−2.39	0.079	0.088	10.95
San Francisco	0.230	0.219	−4.75	0.231	0.250	8.30
Santa Clara	0.162	0.155	−4.42	0.165	0.178	7.98

Notes: Taxes are dollars per vehicle mile and are optimal-flow-weighted means over the county’s hourly profile. The pre-period is 2018–2019 and the post-period is 2021–2022. The fixed-value-of-time calculation holds the county value of time at its 2018 level. The contemporaneous calculation uses one-half of the county median hourly wage in each year. All equilibria impose a private-cost elasticity of -0.1 .

Figure 4 shows the same counterfactual. The levels differ across counties because the estimated congestion curves, network capacities, baseline hourly profiles, and wages differ. San Francisco has the largest dollar-per-mile tax even though its observed detector network is much smaller, reflecting its steep congestion technology and high value of time.

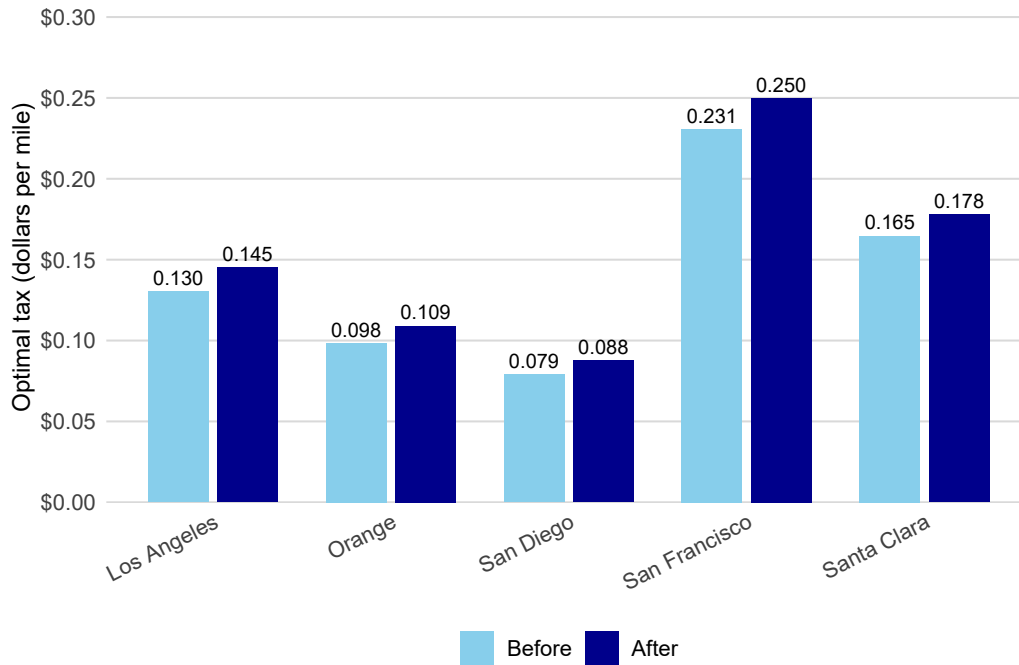
Figure 4. Optimal Congestion Taxes with the Value of Time Fixed



Notes: Optimal-flow-weighted county-period mean taxes in dollars per mile. The value of time is fixed at its 2018 county level.

Allowing wages to change reverses the comparison. As shown in Figure 5, post-period taxes rise by 7.98 percent in Santa Clara, 8.30 percent in San Francisco, and approximately 11 percent in Los Angeles, Orange, and San Diego. The fixed-wage calculation understates the contemporaneous post-period tax by 12.4–13.4 percent.

Figure 5. Optimal Congestion Taxes with Contemporaneous Values of Time



Notes: Optimal-flow-weighted county-period mean taxes in dollars per mile. The value of time equals one-half of the contemporaneous county median hourly wage.

5.2 Why the tax rises

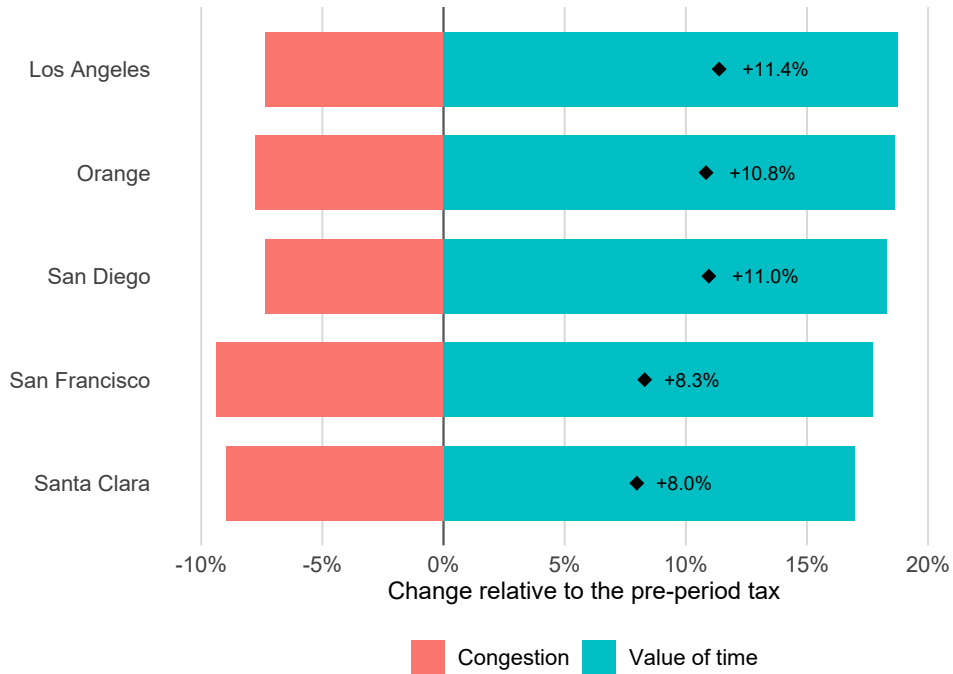
Table 4 and Figure 6 apply Equation (20). The congestion component is negative in every county, ranging from -7.35 to -9.40 percent of the pre-period tax. The value-of-time component is positive and substantially larger, from 16.96 to 18.76 percent. The net effect is therefore an increase in the optimal monetary charge.

Table 4. Decomposition of the Change in the Optimal Tax

County	Congestion component (% of pre tax)	Value-of-time component (% of pre tax)	Total change (% of pre tax)
Los Angeles	−7.38	18.76	11.38
Orange	−7.80	18.63	10.84
San Diego	−7.35	18.30	10.95
San Francisco	−9.40	17.71	8.30
Santa Clara	−8.99	16.96	7.98

Notes: The exact decomposition writes $\tau = VOT \times g$, where g is marginal external delay at the efficient flow. The congestion component holds the value of time at its pre-period level while moving from g_{pre} to g_{post} . The value-of-time component prices post-period delay using the change in the value of time. Components add exactly to the total percentage change, up to rounding.

Figure 6. Decomposition of the Change in the Optimal Tax



Notes: Components are expressed as percentages of the pre-period contemporaneous-wage tax. The congestion and value-of-time components add exactly to the total change.

This result should not be interpreted as evidence that the pandemic increased physical congestion. It did not. Rather, the dollar value attached to a unit of marginal delay rose by

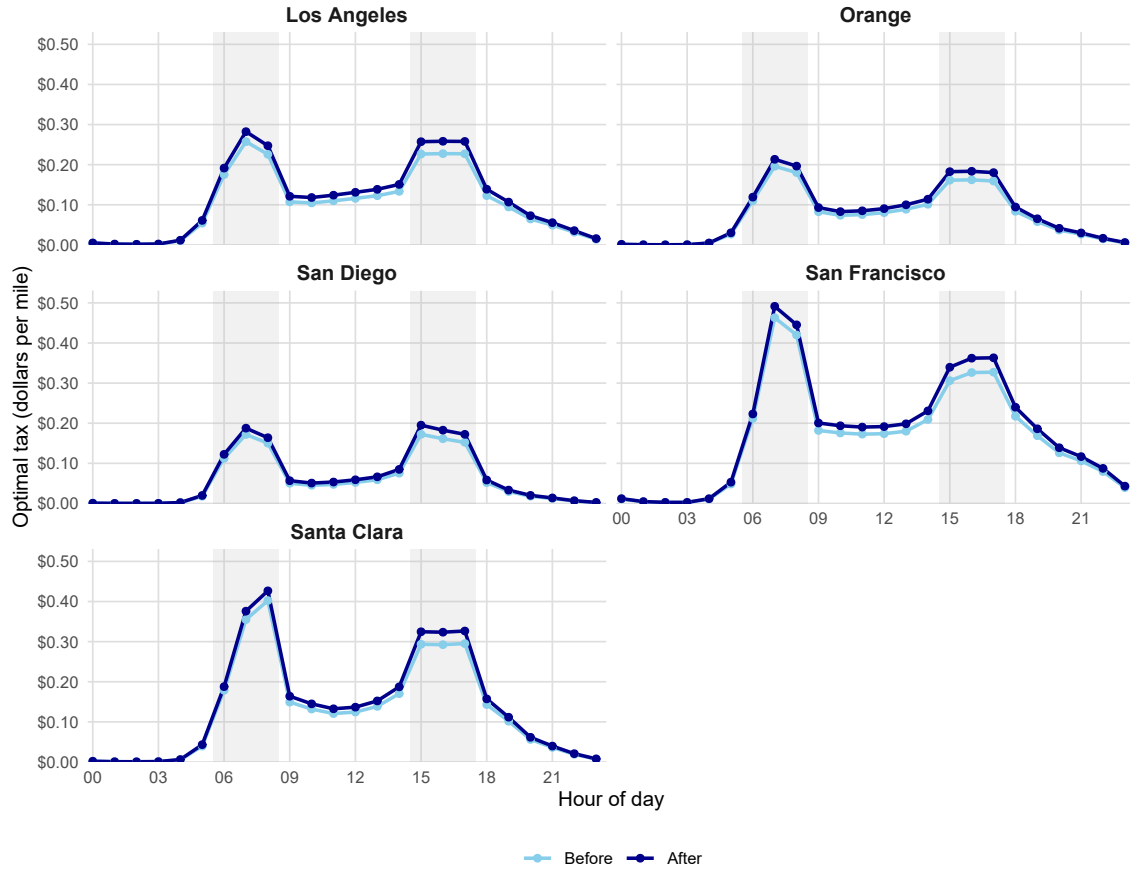
more than the marginal delay itself fell. The sign difference between the fixed- and changing-wage exercises is exactly the distinction between physical congestion and its monetary social cost.

5.3 Time of day

The hourly profile is economically important because the marginal external cost is highly nonlinear in flow. I re-estimate Equation (25) for weekday morning peak hours (6:00–8:59), afternoon peak hours (15:00–17:59), and all other weekday hours. Estimated curvature is 4.240 in the morning, 2.394 in the afternoon, and 2.538 off peak. Within each period, the tax is still solved for every hour rather than assigned as a single period-wide rate.

Figure 7 plots the contemporaneous-wage hourly schedule. Taxes are nonnegative in every county and period: several off-peak hourly rates round to zero, and the model never predicts a negative tax or subsidy. The largest rates occur around the commuting peaks. San Francisco and Santa Clara have especially pronounced morning profiles, while Los Angeles and San Diego exhibit relatively large afternoon rates. The post-period schedule is generally above the pre-period schedule because higher values of time dominate lower marginal delay.

Figure 7. Optimal Congestion Tax over the Weekday Hourly Profile



Notes: County-hour optimal taxes use contemporaneous values of time. Shaded bands denote 6:00–8:59 a.m. and 3:00–5:59 p.m. Morning, afternoon, and off-peak congestion technologies are estimated separately; efficient flow and tax are solved for each hour within those periods.

The period averages in Appendix Figure 9 range from roughly 4 to 14 cents per mile off peak in the pre-period and from 15 to 37 cents per mile during peak periods. With contemporaneous wages, post-period changes are positive in all 15 county-period cells, ranging from 5.85 to 13.54 percent.

5.4 Income heterogeneity and group-specific tax rates

I divide tracts into groups $g \in \{L, H\}$. A tract belongs to the higher-income group when its 2018 median household income is above the household-weighted 80th percentile in its

county; the remaining tracts form the lower-income group. This classification is held fixed over time. Let $\tilde{w}_{gct}$ be the employed-person-weighted median of tract median annual earnings within county-income group gc , divided by 2,080 hours. The group value of time is

$$V_{gct} = \theta \tilde{w}_{gct}, \quad \theta = 0.5. \quad (31)$$

Thus V_{gct} is group g 's value of travel time in dollars per hour in county c and year t ; it is one-half of the group's estimated hourly wage, not its annual household income.

The computation begins by estimating the tract demand equation separately for each group:

$$\ln \widehat{VMT}_{it} = b_g WFH_{it} + \rho_g \ln Population_{it} + \mu_i + \delta_{c(i)t} + e_{it}, \quad i \in \mathcal{I}_{gc}. \quad (32)$$

As in the pooled specification, WFH_{it} is instrumented with the leave-California-out shift-share measure. The estimated coefficients are converted into a county-group-year demand multiplier,

$$m_{gct} = \sum_{i \in \mathcal{I}_{gc}} \omega_{igc}^0 \exp \left\{ \widehat{b}_g (WFH_{it} - WFH_{i0}) + \widehat{\rho}_g [\ln Population_{it} - \ln Population_{i0}] \right\}, \quad (33)$$

$$\omega_{igc}^0 = \frac{\widehat{VMT}_{i0}}{\sum_{k \in \mathcal{I}_{gc}} \widehat{VMT}_{k0}}, \quad 0 \equiv 2018.$$

Thus $m_{gc,2018} = 1$, and subsequent demand changes differ across income groups because both the estimated work-from-home response and the realized tract covariates differ.

I next use the 2018 allocated-VMT share s_{gc}^0 to divide each county-hour's observed flow

between the two groups. With no toll in the calibration year,

$$F_{gch0}^0 = s_{gc}^0 F_{ch0}^{obs}, \quad s_{gc}^0 = \frac{\sum_{i \in \mathcal{I}_{gc}} \widehat{VMT}_{i0}}{\sum_{i \in \mathcal{I}_c} \widehat{VMT}_{i0}}, \quad (34)$$

$$A_{gch0} = F_{gch0}^0 [V_{gc0} T_c(F_{ch0}^{obs})]^\eta, \quad A_{gcht} = A_{gch0} m_{gct}. \quad (35)$$

Equation (35) recovers the group-specific constant-elasticity demand intercept that exactly reproduces the allocated 2018 flow. The multiplier from Equation (33) then shifts that intercept over time. Although demand is allowed to differ by income group, congestion depends on their combined use of the road, $F_{cht} = F_{Lcht} + F_{Hcht}$.

This shared-network structure first implies a common monetary Pigouvian toll. The superscript P denotes the unconstrained Pigouvian allocation; it is a policy label, not the generalized price p used in Equation (7). Thus F_{gcht}^P is efficient flow supplied by group g , F_{cht}^P is their total, and τ_{cht}^P is the common first-best monetary toll in dollars per mile. These objects solve

$$F_{gcht}^P = A_{gcht} [V_{gct} T_c(F_{cht}^P) + \tau_{cht}^P]^{-\eta}, \quad F_{cht}^P = \sum_g F_{gcht}^P, \quad (36)$$

$$\tau_{cht}^P = T'_c(F_{cht}^P) \sum_g V_{gct} F_{gcht}^P. \quad (37)$$

Every additional vehicle mile creates the same physical delay, regardless of which group supplies it, so Equation (37) is the same dollar-per-mile charge for both groups. The additive terms $\tau_{gcht}^{contrib} = T'_c(F_{cht}^P) V_{gct} F_{gcht}^P$ sum to τ_{cht}^P and attribute the external cost to the travelers who bear the added delay. They are contributions to the common toll, not rates levied on each group. Appendix Figure 10 reports this decomposition.

To construct actual group-specific rates, I consider a constrained policy that assigns both groups the same toll measured in travel time, κ_{cht} hours per mile, and converts it into money

using the group value of time:

$$\tau_{gcht}^V = V_{gct} \kappa_{cht}. \quad (38)$$

Here κ_{cht} is the common time-equivalent charge: it adds the same number of travel hours per mile to each group's perceived time cost. It is not itself a dollar toll. Multiplication by V_{gct} converts it into the group-specific monetary rate τ_{gcht}^V , where the superscript V denotes the value-of-time-proportional policy. The generalized monetary price is therefore $V_{gct}[T_c(F_{cht}) + \kappa_{cht}]$.

With the common elasticity η , the constrained optimum and the two group flows are the joint solution to

$$F_{gcht}^\kappa = A_{gcht} \{V_{gct} [T_c(F_{cht}^\kappa) + \kappa_{cht}^*]\}^{-\eta}, \quad F_{cht}^\kappa = \sum_g F_{gcht}^\kappa, \quad (39)$$

$$\kappa_{cht}^* = F_{cht}^\kappa T'_c(F_{cht}^\kappa). \quad (40)$$

The superscript κ on flow denotes equilibrium under this constrained schedule, while κ_{cht}^* is the welfare-maximizing common time charge within that policy class. Because the common Pigouvian policy adds the same dollar toll to both groups whereas the constrained policy adds the same time charge, F_{gcht}^P and F_{gcht}^κ , and therefore their total flows, generally differ. They coincide in the benchmark case $V_{Lct} = V_{Hct}$, when a common time charge is equivalent to a common monetary toll. Substituting the BPR derivative from Equation (2) makes the tax computed in each county-hour explicit:

$$\kappa_{cht}^* = \frac{\alpha_c \beta}{S_c^0} \left(\frac{F_{cht}^\kappa}{K_c} \right)^\beta, \quad (41)$$

$$\tau_{gcht}^V = \frac{V_{gct}}{S_c^0} \alpha_c \beta \left(\frac{F_{cht}^\kappa}{K_c} \right)^\beta. \quad (42)$$

Equations (39)–(42) are solved simultaneously, rather than assigning each group a separate

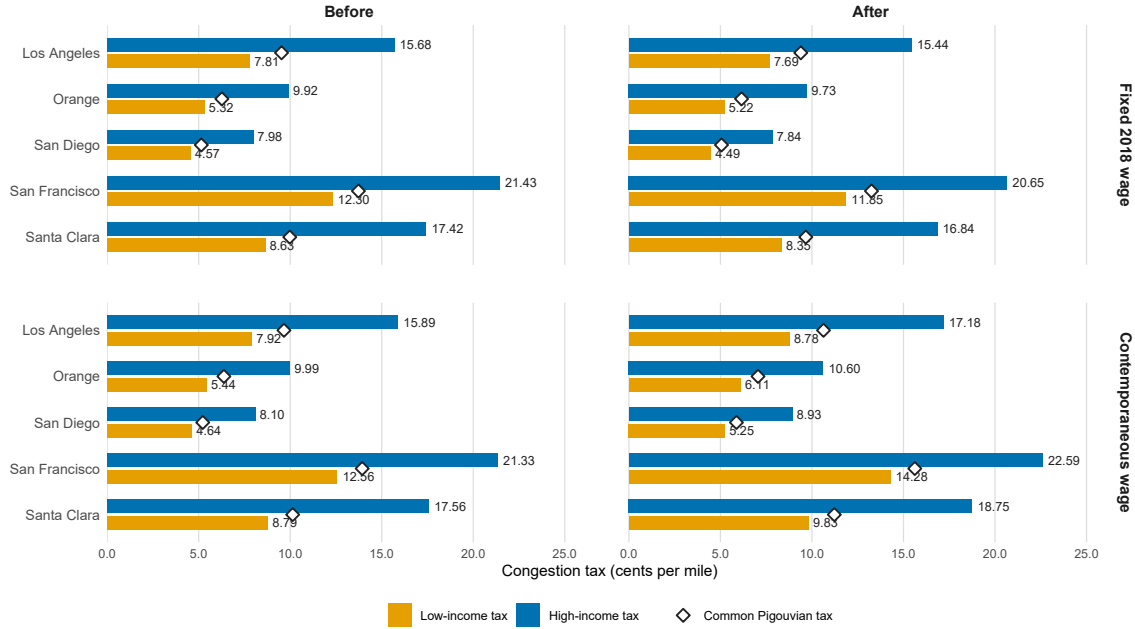
road or holding total traffic fixed. Within the same county-hour, both groups face the same κ_{cht}^* , so

$$\frac{\tau_{Hcht}^V}{\tau_{Lcht}^V} = \frac{V_{Hct}}{V_{Lct}}. \quad (43)$$

This proportional schedule is not the unconstrained first-best common monetary toll in Equation (37); it is the optimum within the restricted class $\tau_g = V_g \kappa$. Appendix E derives both results from welfare.

Figure 8 and Table 5 report the constrained rates. Lower-income tracts account for approximately 77–84 percent of equilibrium flow and therefore for most of the common Pigouvian externality. Nonetheless, their proportional rates are lower because Equation (43) maps the common time charge into dollars through group-specific values of time. In the post-period, higher-income rates are 1.6–2.0 times lower-income rates. For example, the Los Angeles rates are 17.2 and 8.8 cents per mile, respectively. With contemporaneous wages, the post-period rate rises in every county and for both groups: by 10.8–13.7 percent for lower-income tracts and 5.9–10.3 percent for higher-income tracts.

Figure 8. Value-of-Time-Proportional Congestion Taxes by Income Group



Notes: Bars report the constrained group-specific monetary rates $\tau_g = VOT_g \kappa$; the rates and traffic are solved jointly on the shared county network. Diamonds show the common monetary Pigouvian benchmark. The upper panels hold wages at 2018 levels, and the lower panels use contemporaneous wages. Before denotes 2018–2019 and after denotes 2021–2022.

Table 5. Value-of-Time-Proportional Taxes by Income Group

County	Lower-income tracts			Higher-income tracts		
	Pre	Post	Change (%)	Pre	Post	Change (%)
Los Angeles	0.079	0.088	10.84	0.159	0.172	8.14
Orange	0.054	0.061	12.41	0.100	0.106	6.19
San Diego	0.046	0.052	13.17	0.081	0.089	10.25
San Francisco	0.126	0.143	13.73	0.213	0.226	5.91
Santa Clara	0.088	0.098	11.77	0.176	0.187	6.73

Notes: Dollar-per-mile rates under the constrained policy $\tau_g = V_g \kappa$, where the common time-equivalent toll is $\kappa = F^\kappa T'(F^\kappa)$. Group demands and total congestion are solved jointly in every county-hour-year cell, and the table reports group-flow-weighted means. Higher-income tracts are above the household-weighted county 80th percentile of 2018 tract median household income; the classification is held fixed. Group values of time equal one-half of employed-person-weighted median hourly earnings within each county-income group. These are actual group-specific rates under the proportional restriction, not the groups' contributions to the unconstrained common Pigouvian toll.

6 Conclusion

The shift toward remote work changed the demand for freeway travel, but translating that change into a congestion-pricing recommendation requires more than comparing traffic before and after the pandemic. It requires a geographically credible demand measure, a congestion technology expressed in consistent flow and capacity units, and a value of time that converts physical delay into money.

This paper brings those objects together. Fixed pre-pandemic commuting-route exposures allocate observed PeMS station VMT to residential tracts while preserving county totals. A tract-level shift–share design implies that a 10 percentage point increase in remote work reduces allocated freeway VMT by 1.43 percent. A capacity-normalized PPML BPR model yields substantial curvature, and the efficient-flow calculation produces economically meaningful differences over the day and across counties.

The central result is that less congestion does not necessarily imply a lower monetary congestion charge. Holding the value of time fixed, optimal taxes fall after the demand shift. Using contemporaneous wages, they rise. The value placed on delay increases enough to dominate the decline in marginal physical delay. The income analysis also clarifies two objects that have different policy meanings: lower-income tracts account for most of the common marginal external damage because they supply most traffic, while a common time-equivalent charge produces a higher dollar-per-mile rate for the higher-value-of-time group. Distributional conclusions therefore depend on whether the object being compared is responsibility for external delay or the monetary rate paid.

Several limitations define the next steps. Tract freeway VMT is allocated rather than directly observed, the supply relationship is not yet identified by an excluded flow shifter, and the shift–share inference is based on a short time dimension. The income analysis also uses residential tract income as a proxy for the travelers using the network. These limitations call

for route-validation exercises, exposure-robust inference, and richer information on traveler composition. They do not alter the paper’s accounting point: an empirically useful congestion price must value the delay generated by the efficient traffic flow using the value of time of the travelers affected.

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A Derivation of Private, External, and Social Costs

This appendix derives the central cost equations from primitives. Let F be aggregate freeway VMT per hour, $T(F)$ travel time in hours per mile, and V the value of time in dollars per hour. Aggregate travel-time cost is

$$C(F) = VFT(F), \quad (44)$$

measured in dollars per hour. A nonatomic driver treats $T(F)$ as given. Her private cost of one additional vehicle mile is therefore

$$PMC(F) = VT(F). \quad (45)$$

The planner recognizes that an additional unit of flow changes the travel time experienced by every unit already on the road:

$$MSC(F) = \frac{dC(F)}{dF} \quad (46)$$

$$= V [T(F) + FT'(F)] \quad (47)$$

$$= \underbrace{VT(F)}_{PMC(F)} + \underbrace{VFT'(F)}_{MECC(F)}. \quad (48)$$

This is the origin of Equations (4) and (5): the first term is the entrant's own time cost and the second is the time cost imposed on all existing flow.

For the capacity-normalized BPR function,

$$T(F) = \frac{1}{S_0} [1 + \alpha(F/K)^\beta], \quad (49)$$

the derivative is

$$T'(F) = \frac{\alpha\beta}{S^0 K} (F/K)^{\beta-1}. \quad (50)$$

Multiplying by F gives

$$FT'(F) = \frac{\alpha\beta}{S^0} (F/K)^\beta. \quad (51)$$

Therefore,

$$PMC(F) = \frac{V}{S^0} [1 + \alpha(F/K)^\beta], \quad (52)$$

$$MECC(F) = \frac{V}{S^0} \alpha\beta (F/K)^\beta, \quad (53)$$

$$MSC(F) = \frac{V}{S^0} [1 + \alpha(1 + \beta)(F/K)^\beta]. \quad (54)$$

All three expressions are dollars per mile. Capacity is essential for interpreting the flow ratio, while free-flow speed determines the conversion from a time index to hours per mile.

B Efficient Flow, Optimal Toll, and Pass-Through

Let $P(F)$ denote inverse demand in dollars per mile. Net social surplus is

$$W(F) = \int_0^F P(x) dx - VFT(F). \quad (55)$$

The first-order condition is

$$P(F^*) = V[T(F^*) + F^*T'(F^*)] = MSC(F^*). \quad (56)$$

Private equilibrium under a tax τ satisfies $P(F) = VT(F) + \tau$. Evaluating this condition at F^* and comparing it with Equation (56) proves that

$$\tau^* = VF^*T'(F^*) = MECC(F^*). \quad (57)$$

For the constant-elasticity demand $F = A[PMC(F) + \tau]^{-\eta}$, log differentiation yields

$$\frac{1}{F} \frac{dF}{d\tau} = -\eta \frac{PMC'(F)dF/d\tau + 1}{PMC(F) + \tau}, \quad (58)$$

which rearranges to Equation (13). Because $PMC'(F) = VT'(F)$, the optimal tax also satisfies $\tau^* = F^* PMC'(F^*)$. Substitution into Equation (15) gives

$$\left. \frac{dp}{d\tau} \right|_{\tau=\tau^*} = \frac{PMC(F^*) + MECC(F^*)}{PMC(F^*) + MECC(F^*) + \eta MECC(F^*)} = \frac{MSC(F^*)}{MSC(F^*) + \eta MECC(F^*)}. \quad (59)$$

The expression lies strictly between zero and one whenever congestion is present. It characterizes the marginal incidence of the optimal toll; it is not a separate rule for selecting its level.

C Tract Freeway VMT and the Shift–Share Design

Equation (22) combines two distinct sources of variation. The numerator $\sum_j \bar{N}_{ij}^{pre} A_{ijs}$ measures the pre-pandemic commuting exposure of origin tract i to station s . The denominator repeats the same calculation over every candidate origin k in the county. For any matched station, summing that ratio over i therefore gives one. The second bracket, $\sum_{h \in t} Flow_{sh} L_s$, is the station's observed annual VMT contribution. Summing Equation (22) over origin tracts consequently reproduces total PeMS VMT for the county, as stated in Equation (23).

The indices i and j uniquely identify an origin–destination pair; no redundant county

index is needed inside the commuting-pair sums. County enters only through the origin-tract set $\mathcal{I}_{c(i)}$ and the station set $\mathcal{S}_{c(i)}$. Because the LODES counts and route indicators are fixed in the pre-period, they determine only how observed station VMT is distributed across tracts. Year-to-year traffic movement comes from PeMS. The resulting $\widehat{VMT}_{it}$ should therefore be interpreted as exposure-allocated freeway VMT associated with tract i ; changes in unobserved route choice within an origin tract are not measured.

The first stage corresponding to Equation (26) is

$$WFH_{it} = \xi_i^F + \delta_{c(i)t}^F + \pi Z_{it} + \gamma_F \ln Population_{it} + X'_{it} \lambda + u_{it}. \quad (60)$$

The leave-California-out construction limits mechanical correlation between local remote-work measurement and the national industry shocks. The exclusion restriction may still fail if national industry shocks affect tract traffic through channels other than remote work that vary within county-year in proportion to 2019 industry exposure. Sector-specific employment or income shocks are the main concern. The demographic-control specification and industry-level shock diagnostics are therefore important robustness checks.

D Time-of-Day Congestion Technology

For time period $r \in \{morning, afternoon, offpeak\}$, I estimate

$$E[D_{cht} \mid F_{cht}, c, r] = \alpha_{cr} \left(\frac{F_{cht}}{K_c} \right)^{\beta_r}. \quad (61)$$

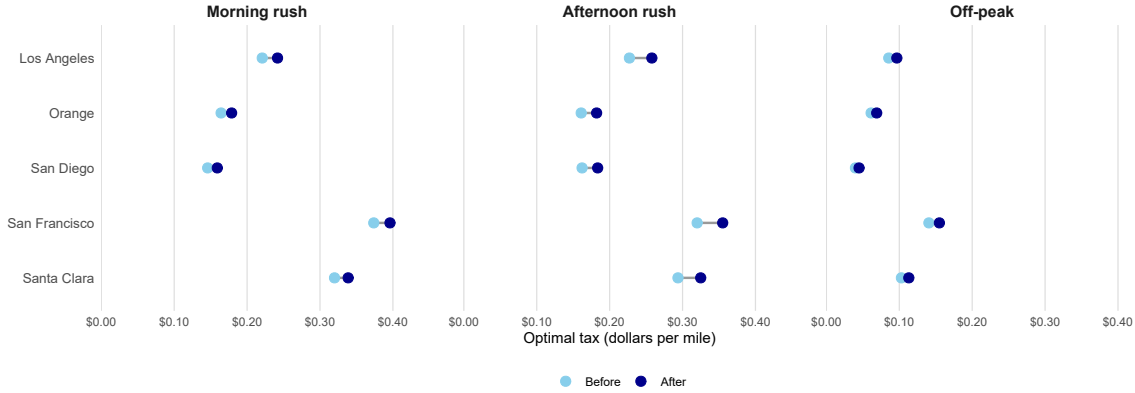
Capacity and free-flow speed remain county-specific engineering inputs, while the BPR scale and curvature can differ across periods. This allows the same flow-capacity ratio to produce different expected delay at different times of day, reflecting systematic changes in bottlenecks and network conditions.

Table 6. Congestion Curvature by Time of Day

Period	Hours	$\hat{\beta}$	HC1 standard error
Morning peak	6:00–8:59	4.240	0.040
Afternoon peak	15:00–17:59	2.394	0.089
Off-peak	Other weekday hours	2.538	0.021

Notes: Each row comes from a separate weekday PPML BPR regression with a common curvature parameter within the period and county-specific scale parameters. The morning and afternoon regressions each use 15,645 observations; the off-peak regression uses 93,870. The period-specific congestion curve is then used to solve the hourly efficient flow and tax. Standard errors are heteroskedasticity robust.

Figure 9. Optimal Congestion Taxes by Time Period



Notes: Optimal-flow-weighted period means use contemporaneous values of time. Morning peak is 6:00–8:59, afternoon peak is 15:00–17:59, and off-peak contains all other weekday hours.

E Welfare Derivation for Income-Group Taxes

Suppress county-hour-year subscripts and let groups be indexed by $g \in \{L, H\}$. Group g has inverse demand $P_g(F_g)$, value of time V_g , and flow F_g ; total flow is $F = \sum_g F_g$. Welfare net of travel-time cost is

$$W(F_L, F_H) = \sum_g \int_0^{F_g} P_g(x) dx - T(F) \sum_g V_g F_g. \quad (62)$$

Define $M \equiv \sum_g V_g F_g$, the value-of-time-weighted flow exposed to congestion. The welfare derivative for one additional vehicle mile supplied by group q is

$$\frac{\partial W}{\partial F_q} = P_q(F_q) - \underbrace{V_q T(F)}_{\text{private time cost}} - \underbrace{T'(F) M}_{\text{external congestion cost}}. \quad (63)$$

The private term differs with the entrant's value of time. The external term does not depend on the entrant's identity: one additional vehicle mile changes travel time for all existing users. A common monetary toll yields the private condition $P_q(F_q) = V_q T(F) + \tau^P$. Comparing it with Equation (63) gives

$$\tau^P = T'(F^P) M^P = T'(F^P) \sum_g V_g F_g^P, \quad (64)$$

which proves Equation (37). For accounting purposes, this common toll can be written as the sum

$$\tau^P = \sum_g \tau_g^{\text{contrib}}, \quad \tau_g^{\text{contrib}} = T'(F^P) V_g F_g^P. \quad (65)$$

Each contribution measures the portion of marginal external damage borne by group g . It is not the tax rate paid by an entrant from that group.

I next restrict policy to $\tau_g = V_g \kappa$, where κ is a common time-equivalent toll in hours per

mile. The private equilibrium condition and constant-elasticity group demand are

$$P_g(F_g) = V_g[T(F) + \kappa], \quad F_g = A_g V_g^{-\eta} [T(F) + \kappa]^{-\eta}. \quad (66)$$

Because both groups share the same elasticity η , their flow shares do not change with a marginal change in κ within a county-hour-year cell:

$$\frac{F_g}{F} = \frac{A_g V_g^{-\eta}}{\sum_{\ell} A_{\ell} V_{\ell}^{-\eta}}, \quad \frac{dF_g}{d\kappa} = \frac{F_g}{F} \frac{dF}{d\kappa}. \quad (67)$$

Differentiating Equation (62) along the equilibrium response to κ gives

$$\begin{aligned} \frac{dW}{d\kappa} &= \sum_g P_g(F_g) \frac{dF_g}{d\kappa} - T(F) \sum_g V_g \frac{dF_g}{d\kappa} - T'(F) M \frac{dF}{d\kappa} \\ &= \kappa \sum_g V_g \frac{dF_g}{d\kappa} - T'(F) M \frac{dF}{d\kappa} \\ &= M \frac{dF}{d\kappa} \left[\frac{\kappa}{F} - T'(F) \right]. \end{aligned} \quad (68)$$

The last equality uses Equation (67), which implies $\sum_g V_g (dF_g/d\kappa) = (M/F)(dF/d\kappa)$. Because $M > 0$ and $dF/d\kappa < 0$, welfare rises with the time toll below $FT'(F)$ and falls above it. The constrained optimum therefore satisfies

$$\kappa^* = F^{\kappa} T'(F^{\kappa}). \quad (69)$$

This establishes Equation (40). Under the BPR technology,

$$\kappa^* = \frac{\alpha\beta}{S^0} \left(\frac{F^{\kappa}}{K} \right)^{\beta}, \quad \tau_g^V = V_g \kappa^*. \quad (70)$$

The numerical calculation can be written as a one-dimensional fixed-point problem. For

each county-hour-year cell, I solve the positive root of

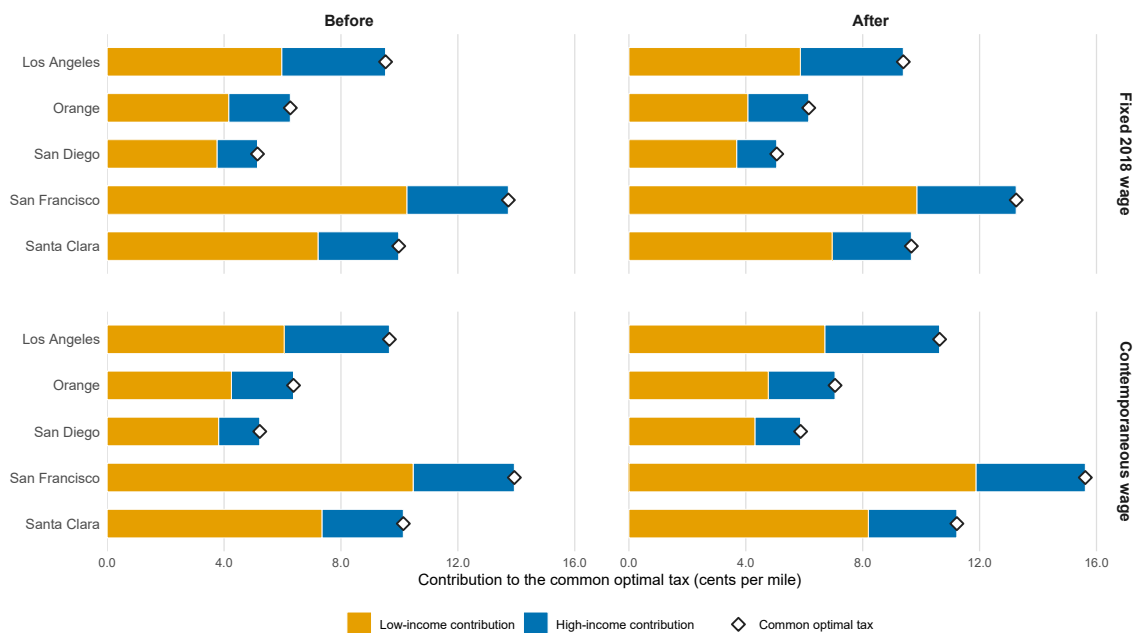
$$\Phi(F) = \sum_g A_g \{V_g[T(F) + FT'(F)]\}^{-\eta} - F = 0. \quad (71)$$

I then recover $\kappa^* = F^\kappa T'(F^\kappa)$, each group flow from Equation (66), and each monetary rate from $\tau_g^V = V_g \kappa^*$. The reported pre- and post-period group rates are group-flow-weighted means over the 168-hour weekly profile and the two years in each period:

$$\bar{\tau}_{gc,r}^V = \frac{\sum_{t \in r} \sum_{h \in \mathcal{H}} \tau_{gcht}^V F_{gcht}^\kappa}{\sum_{t \in r} \sum_{h \in \mathcal{H}} F_{gcht}^\kappa}, \quad r \in \{pre, post\}. \quad (72)$$

The fixed-wage exercise holds V_g at its 2018 group value in every year; the contemporaneous exercise updates it using that group's year-specific wage. If group elasticities differ, Equation (67) no longer holds, and the constrained optimum must instead be obtained from the unrestricted derivative in the second line of Equation (68).

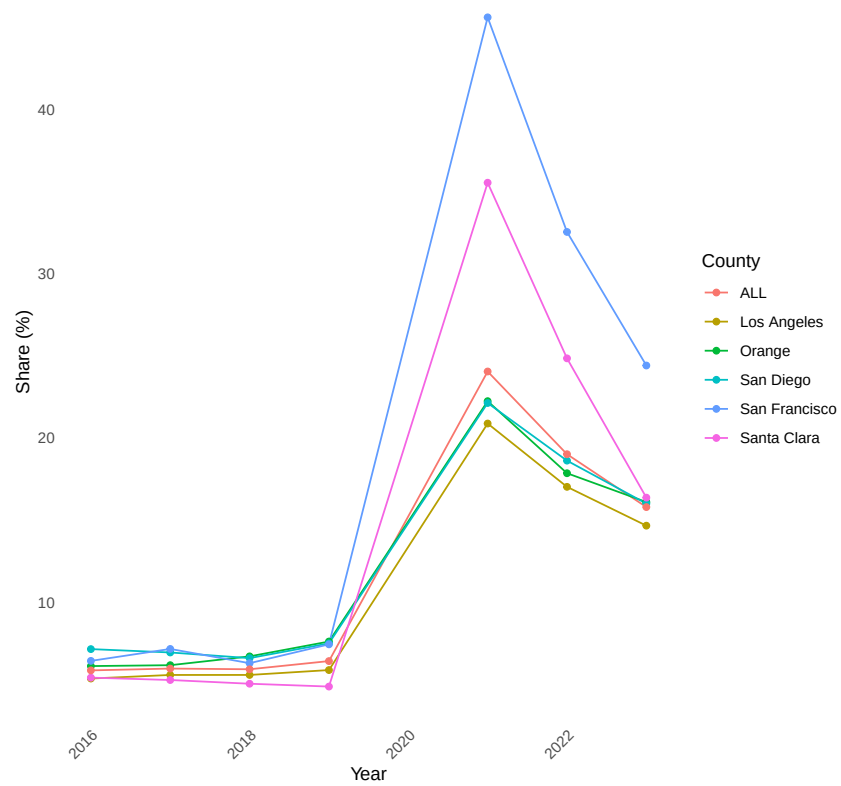
Figure 10. Income-Group Contributions to the Common Optimal Tax



Notes: Each component equals $T'(F^*)V_g F_g^*$ and the two components sum to the common monetary Pigouvian tax shown by the diamond. The components describe external-delay damages attributable to each group; they are not separate toll rates. The upper panels hold wages at 2018 levels, and the lower panels use contemporaneous wages. Before denotes 2018–2019 and after denotes 2021–2022.

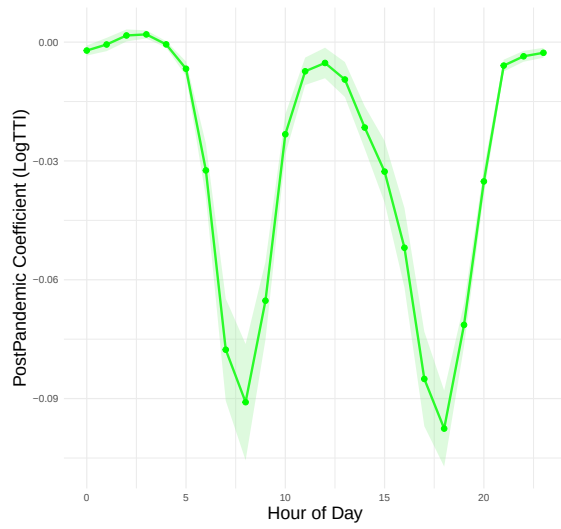
F Additional Descriptive Figures

Figure 11. Work-from-Home Shares by County

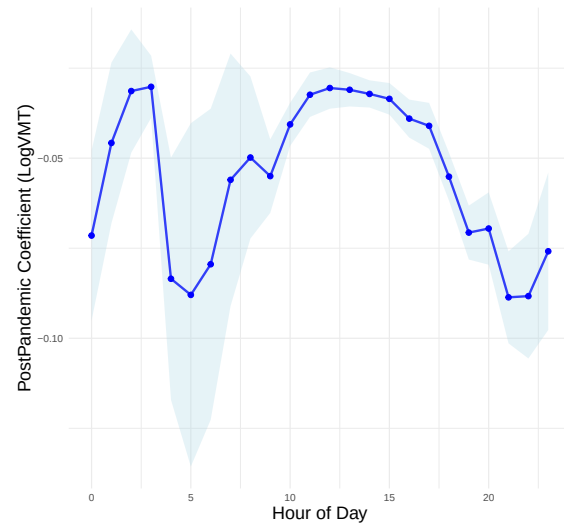


Notes: ACS work-from-home shares for the five study counties.

Figure 12. Hourly Changes in Traffic Conditions



(a) Travel Time Index



(b) Vehicle Miles Traveled

Notes: Hourly PeMS traffic profiles before and after the pandemic demand shift.